

Task 1. EDA

Objective: perform exploratory data analysis on real cryptocurrency derivatives data from Binance and Bybit to uncover patterns in trading activity, liquidity, and liquidation events. The main objective of this exercise is to get familiar with the data.

Data: <https://drive.google.com/file/d/1FpLBVKViVhvrFXw1vgBL743Y1IFWDwFD/view?pli=1>

Deliverables: Jupyter Notebook

- Clean, documented code with markdown explanations
- Summary table of key findings per tier completed
- Executive summary with actionable insights and limitations of the analysis & data caveats

Assessment:

- Technical execution: data handling, code quality, visualization, etc.
- Analytical depth: pattern discovery, hypothesis generation, critical reflection, etc.
- Communication & Deliverables: notebook structure, executive summary, reproducibility

Task 1.1 Foundational EDA

Focus: Data understanding & basic visualization

1. Data Quality Check
 - a. Missing values, duplicates, timestamp consistency
 - b. Compare data frequency across streams (trades vs BBO vs liquidations)
 - c. ...
2. Univariate Analysis
 - a. Distribution of trade sizes, prices, liquidation volumes
 - b. Time-of-day patterns: when do most liquidations occur?
 - c. ...
3. Basic Cross-Exchange Comparison
 - a. Plot liquidation volume by hour for Binance vs Bybit
 - b. Which exchange shows higher liquidation frequency?
 - c. ...

Task 1.2 Intermediate Analysis

Focus: Relationships, market microstructure, hypothesis testing

1. Broad market EDA
 - a. spread distributions,
 - b. trade stats (size, taker/maker, frequency)
 - c. ...
2. ML EDA
 - a. non-linearities,
 - b. regime structure (goal is to come up with meaningful for EDA market regimes),

- c. signal-to-noise on candidate targets,
- d. target stationarity,
- e. outlier structure,
- f. Autocorrelation (on different scales),
- g. ...

Task 1.3 Advanced Exploration (no hints here)

Focus: Feature engineering, predictive signals, critical thinking

1. Orderbook metrics & price impact
2. Side imbalance and liquidations analysis
3. Liquidation heatmap construction
4. Cross-Exchange arbitrage signals
5. ...